§3.2 Growth of Functions

Big-O and related notation

Q: What problem are solving?

A: How can we compare algorithms?

Example
What is the 'best' way to multiply natural numbers.

i.e 136×47

"Best" ways:
- Fastest
- Least Memory
- Least Training
- Least likely to make a mistake
- etc.

Usually, "whats the fastest way" (Fastest, least memory, least training)

Problem 1 to avoid:
For small instances, the fastest solution is pre-calculate and use a look-up table.

Problem 2
At least in timing, how do we make the race fair.

- By hand
- By calculator

vs
- By supercomputer

To solve problem 1

Only compare algorithms when they solve "big enough" problems

For Problem 2

If the race would switch (winners) by using a faster computer for one team, call it a tie.

Big-O notation

For two functions, f and g with domain $\mathbb{Z}$ or $\mathbb{R}$ (or $\mathbb{N}, \mathbb{Q}$, somewhere where we can head to infinity)

and codomain $\mathbb{R}$ (or some subset)

We say $f(x)$ is $\mathcal{O}(g(x))$

We read this as: "f of x is big-O of g of x"

If there exists witnesses $C > 0$ and k
$|f(x)| \leq C\ |g(x)|$ whenever $x > k$ (sometimes $x \geq k$)

if f and g measure algorithms

k lets us throw away small exmaples,

C lets us run the first algorithm on faster hardware.

Conceptually
The algorithm represented by f is not a clear loser to the one represented by g.

If C does not exist, then no amount of hardware speedup can help C for large problems.

Bad Example of an algorithm for multiplication % To compute $m \times n$

```
s:= 0

for i from 1 to n do
  s:= s + m

next i
return s
```

This scales really badly.

Example

Suppose $f(n)$ is $1^5 + 2^5 + 3^5 + ... + n^5$

Note: $f : \mathbb{N} \to \mathbb{R}$

Observe that
$|f(n)| = |1^5 + 2^5 + 3^5 + ...n^5|$
$\leq |1^5| + |2^5| + |3^5| + ... + |n^5|$
$\leq n^5 + n^5 + n^5 + ...n^5$ (as long as $n \geq 1$)
$= n^6 = 1 * |n^6|$

Thus $f(n)$ is $\mathcal{O}(n^6)$

Theorem 1
If $g(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x^1 + a_0 x^0$
i.e $g(x)$ is a polynomial of degree at most n
then $g(x)$ is $\mathcal{O}(x^n)$, even if $a_n = 0$

In this case:
$|g(x)| = |a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x^1 + a_0$
$\leq |a_n x^n| + |a_{n-1}x^{n-1}| + ... + |a_1 x^1 + a_0|$
$= x^n \left(|a_n| + |a_{n-1}\frac{}{}x| + |a_{n-2}\frac{}{}x^1| + ... + |a_1\frac{}{}x^{n-1}| + |a_0\frac{}{}x^{n-2}| \right)$ (factor out the $|x^n|$)

As long as $|x| > 1, |a_n \frac{1}{|} x| \leq |a_{n-1}|$ since $a_{n-i} = 0$ is possible.

$\leq |x^n|(|a_n| + |a_{n-1}| + ... + |a_1| + |a_0|)$ (Think C)

the $|x^n|$ is $h(x)$

So $g(x)$ is $\mathcal{O}(x^n)$

with witnesses $k = 1$ and :
$C = |a_n| + |a_{n-1}| + ... + |a_1| + |a_0|$

The witnesses are not unique, once we have witnesses, C and k, we can use any bigger C or any bigger k, and sometimes using a bigger k lets us a smaller C and vice-versa.